ACM/IDS 104 APPLIED LINEAR ALGEBRA
PRACTICE PROBLEMS FOR LECTURE 10: SOLUTIONS

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Problem 10A. Least Squares via Orthogonal Projection
In Lecture 10, we discussed the geometric interpretation of least squares: to find a least squares solution of the system $Ax = b$, we need to orthogonally project the vector b onto $\operatorname{im} A$, and then solve the system $Ax = \operatorname{pr}_{\operatorname{im} A} b$ in the ordinary (exact) sense. Let's revisit the linear system from Practice Problem 7A:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \quad \text{and} \quad b = \begin{bmatrix} 3 \\ 3 \\ 9 \end{bmatrix}. \tag{1}$$

(a) Find an orthonormal basis of $\operatorname{im} A$.
(b) Find the orthogonal projection of b onto $\operatorname{im} A$.
(c) Find the least squares solutions of $Ax = b$ by solving $Ax = \operatorname{pr}_{\operatorname{im} A} b$.

Solution:
(a) As we discussed in Practice Problem 7A (and Problem 5 of PS1), $\operatorname{rank} A = 2$. Therefore, $\dim \operatorname{im} A = 2$, and the first two columns of A,

$$a_1 = \begin{bmatrix} 1 \\ 4 \\ 7 \end{bmatrix} \quad \text{and} \quad a_2 = \begin{bmatrix} 2 \\ 5 \\ 8 \end{bmatrix}, \tag{2}$$

form a basis of $\operatorname{im} A$. To find an orthonormal basis $\{u_1, u_2\}$ of $\operatorname{im} A$, let's apply the Gram-Schmidt process to a_1 and a_2:

$$\begin{aligned} v_1 &= a_1 = \begin{bmatrix} 1 \\ 4 \\ 7 \end{bmatrix} \quad \Rightarrow \quad u_1 = \frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{66}} \begin{bmatrix} 1 \\ 4 \\ 7 \end{bmatrix}, \\ v_2 &= a_2 - \frac{\langle a_2, v_1 \rangle}{\|v_1\|^2} v_1 = \begin{bmatrix} 2 \\ 5 \\ 8 \end{bmatrix} - \frac{78}{66} \begin{bmatrix} 1 \\ 4 \\ 7 \end{bmatrix} = \begin{bmatrix} \frac{9}{11} \\ \frac{3}{11} \\ -\frac{3}{11} \end{bmatrix} \quad \Rightarrow \quad u_2 = \frac{v_2}{\|v_2\|} = \frac{1}{\sqrt{11}} \begin{bmatrix} 3 \\ 1 \\ -1 \end{bmatrix}. \end{aligned} \tag{3}$$

(b) The coordinates of the orthogonal projection $\operatorname{pr}_{\operatorname{im} A} b$ in the orthonormal basis u_1, u_2 of $\operatorname{im} A$ are simply the inner products of b with the basis vectors:

$$\begin{aligned} \operatorname{pr}_{\operatorname{im} A} &= \alpha_1 u_1 + \alpha_2 u_2, \\ \alpha_1 &= \langle b, u_1 \rangle = \frac{78}{\sqrt{66}}, \\ \alpha_2 &= \langle b, u_2 \rangle = \frac{3}{\sqrt{11}}, \end{aligned} \quad \Rightarrow \quad \operatorname{pr}_{\operatorname{im} A} = \frac{13}{11} \begin{bmatrix} 1 \\ 4 \\ 7 \end{bmatrix} + \frac{3}{11} \begin{bmatrix} 3 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 8 \end{bmatrix}. \tag{4}$$

(c) To find the least squares solutions of $Ax = b$, it remains to solve the following system in the ordinary (exact) sense:

$$Ax = \operatorname{pr}_{\operatorname{im} A} b \quad \Leftrightarrow \quad \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 8 \end{bmatrix}. \tag{5}$$

In Practice Problem 7A, we found that $\ker A$ is one-dimensional and is spanned by the vector $k = [1, -2, 1]^T$. A particular solution of (5) is $x^* = [0, 1, 0]^T$. Therefore, the set of all least squares solutions of $Ax = b$ is given by

$$x = x^* + \ker A = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + \alpha \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}, \quad \text{where } \alpha \in \mathbb{R}. \tag{6}$$

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Problem 10B. Approximation of Functions by Polynomials

In Lecture 10, we discussed how to approximate a function $f(x)$ by a polynomial $p(x) \in \mathcal{P}^{(n)}[a,b]$ of degree at most n on an interval $[a,b]$. All we need to do is to project $f(x)$ onto $\mathcal{P}^{(n)}[a,b]$ orthogonally with respect to the L^2 inner product, $p(x) = \mathrm{pr}_{\mathcal{P}^{(n)}[a,b]} f(x)$. The polynomial $p(x)$ is then the closest (in the L^2 sense) polynomial in $\mathcal{P}^{(n)}[a,b]$ to the function $f(x)$.

Let $f(x) = e^x$ and $a > 0$ be a positive real number.

(a) Find the best L^2-approximation $p_a(x) \in \mathcal{P}^{(2)}[-a,a]$ to $f(x)$.
(b) Compute the polynomial limit $t(x) = \lim_{a\to 0} p_a(x)$. Does it remind you anything?

Solution:

(a) Our plan for solving this problem is to construct an orthonormal basis $\{Q_0(x), Q_1(x), Q_2(x)\}$ of $\mathcal{P}^{(2)}[-a,a]$, and then compute $p_a(x) = \alpha_0 Q_0(x) + \alpha_1 Q_1(x) + \alpha_2 Q_2(x)$, where the coordinates $\alpha_0, \alpha_1, \alpha_2$ are the inner products of $f(x)$ with $Q_0(x), Q_1(x), Q_2(x)$, respectively. To execute this plan, we can use some of the formulas derived in Lecture 10, but instead, for illustrative purposes, we will do everything from scratch.

A natural basis of $\mathcal{P}^{(n)}[-a,a]$ is $\{1, x, x^2\}$. To orthogonalize it, we will use the Gram-Schmidt process:

$$\begin{aligned}
q_0(x) &= 1,\\
q_1(x) &= x - \frac{\langle x, q_0\rangle}{\|q_0\|^2} q_0 = x - \frac{\int_{-a}^{a} x dx}{\int_{-a}^{a} 1 dx} = x - \frac{0}{2a} = x,\\
q_2(x) &= x^2 - \frac{\langle x^2, q_0\rangle}{\|q_0\|^2} q_0 - \frac{\langle x^2, q_1\rangle}{\|q_1\|^2} q_1 = x^2 - \frac{\int_{-a}^{a} x^2 dx}{\int_{-a}^{a} 1 dx} - \frac{\int_{-a}^{a} x^3 dx}{\int_{-a}^{a} x^2 dx} x = x^2 - \frac{\frac{2a^3}{3}}{2a} = x^2 - \frac{a^2}{3}.
\end{aligned} \tag{7}$$

Next, to construct an orthonormal basis, we normalize $q_0(x), q_1(x)$, and $q_2(x)$:

$$\begin{aligned}
Q_0(x) &= \frac{q_0(x)}{\|q_0\|} = \frac{1}{\sqrt{\int_{-a}^{a} 1 dx}} = \frac{1}{\sqrt{2a}},\\
Q_1(x) &= \frac{q_1(x)}{\|q_1\|} = \frac{x}{\sqrt{\int_{-a}^{a} x^2 dx}} = \sqrt{\frac{3}{2a^3}} x,\\
Q_2(x) &= \frac{q_2(x)}{\|q_2\|} = \frac{x^2 - \frac{a^2}{3}}{\sqrt{\int_{-a}^{a} (x^2 - \frac{a^2}{3})^2 dx}} = \frac{x^2 - \frac{a^2}{3}}{\sqrt{\frac{8a^5}{45}}} = \sqrt{\frac{45}{8a^5}} (x^2 - \frac{a^2}{3}).
\end{aligned} \tag{8}$$

The best L^2-approximation is given by

$$p_a(x) = \alpha_0 Q_0(x) + \alpha_1 Q_1(x) + \alpha_2 Q_2(x), \tag{9}$$

where the coordinates (the integrals can be computed using the integration by parts[1])

$$\begin{aligned}
\alpha_0 &= \langle f, Q_0\rangle = \int_{-a}^{a} e^x \frac{1}{\sqrt{2a}} dx = \frac{e^a - e^{-a}}{\sqrt{2a}},\\
\alpha_1 &= \langle f, Q_1\rangle = \int_{-a}^{a} e^x \sqrt{\frac{3}{2a^3}} x dx = \sqrt{\frac{3}{2a^3}} \left(e^a(a-1) + e^{-a}(a+1)\right),\\
\alpha_2 &= \langle f, Q_2\rangle = \int_{-a}^{a} e^x \sqrt{\frac{45}{8a^5}} (x^2 - \frac{a^2}{3}) dx = \sqrt{\frac{45}{8a^5}} \left(e^a \left(\frac{2}{3}a^2 - 2a + 2\right) - e^{-a}\left(\frac{2}{3}a^2 + 2a + 2\right)\right)
\end{aligned} \tag{10}$$

Finally,

$$p_a(x) = \frac{e^a - e^{-a}}{2a} + \frac{3}{2a^3}\left(e^a(a-1) + e^{-a}(a+1)\right) x + \frac{45}{8a^5}\left(e^a\left(\frac{2}{3}a^2 - 2a + 2\right) - e^{-a}\left(\frac{2}{3}a^2 + 2a + 2\right)\right)\left(x^2 - \frac{a^2}{3}\right) \tag{11}$$

[1] Or using WolframAlpha :)

(b) Now, let's have some fun and find the limit of $p_a(x)$ as $a \to 0$. Using the Taylor expansion of the exponential function about $a = 0$, we have:

$$\begin{aligned}
&\frac{e^a - e^{-a}}{2a} = \frac{(1 + a + O(a^2)) - (1 - a + O(a^2))}{2a} = 1 + O(a) \to 1, \quad \text{as } a \to 0, \\
&\frac{3}{2a^3}\left(e^a(a-1) + e^{-a}(a+1)\right) \\
&= \frac{3}{2a^3}\left(\left(1 + a + \frac{a^2}{2} + \frac{a^3}{6} + O(a^4)\right)(a-1) + \left(1 - a + \frac{a^2}{2} - \frac{a^3}{6} + O(a^4)\right)(a+1)\right) \\
&= \frac{3}{2a^3}\left(a\left(2 + a^2 + O(a^4)\right) - 2a - \frac{a^3}{3} + O(a^4)\right) = \frac{3}{2a^3}\left(\frac{2a^3}{3} + O(a^4)\right) \to 1, \quad \text{as } a \to 0.
\end{aligned} \tag{12}$$

To find the limit of the last term, we need to Taylor-expand e^a up terms of order 5[2]:

$$\begin{aligned}
&\frac{45}{8a^5}\left(e^a\left(\frac{2}{3}a^2 - 2a + 2\right) - e^{-a}\left(\frac{2}{3}a^2 + 2a + 2\right)\right) \\
&= \frac{45}{8a^5}\left(\left(1 + a + \frac{a^2}{2} + \frac{a^3}{6} + \frac{a^4}{24} + \frac{a^5}{120} + O(a^6)\right)\left(\frac{2}{3}a^2 - 2a + 2\right)\right. \\
&\left. - \left(1 - a + \frac{a^2}{2} - \frac{a^3}{6} + \frac{a^4}{24} - \frac{a^5}{120} + O(a^6)\right)\left(\frac{2}{3}a^2 + 2a + 2\right)\right) \\
&= \frac{45}{8a^5}\left(\frac{2a^2}{3}\left(2a + \frac{a^3}{3} + \frac{a^5}{60} + +O(a^6)\right) - 2a\left(2 + a^2 + \frac{a^4}{12} + O(a^6)\right) + 2\left(2a + \frac{a^3}{3} + \frac{a^5}{60} + +O(a^6)\right)\right) \\
&= \frac{45}{8a^5}\left(\frac{4a^3}{3} + \frac{2a^5}{9} + O(a^6) - 4a - 2a^3 - \frac{a^5}{6} + O(a^6) + 4a + \frac{2a^3}{3} + \frac{a^5}{30} + O(a^6)\right) \\
&= \frac{45}{8a^5}\left(\frac{2a^5}{9} - \frac{a^5}{6} + \frac{a^5}{30} + O(a^6)\right) = \frac{45}{8a^5}\left(\frac{8a^5}{90} + O(a^6)\right) \to \frac{1}{2}, \quad \text{as } a \to 0.
\end{aligned} \tag{13}$$

Therefore,

$$t(x) = \lim_{a \to 0} p_a(x) = 1 + x + \frac{1}{2}x^2, \tag{14}$$

which is simply the degree-2 Taylor polynomial of $f(x) = e^x$ at $x = 0$.

Problem 10C. Fourier Approximations

This is a bonus problem for Linear Algebra and Analysis lovers! :) As we know from Lecture 10 and from Practice Problem 10B, we can approximate functions by polynomials by projecting them orthogonally onto the appropriate polynomial vector spaces. Instead of polynomials, we can also use trigonometric functions and vector spaces. This idea leads to what is called *Fourier Analysis*, an area of analysis often used in applications. In this problem, we will just get a flavor for it.

(a) Show that, for each $n \geq 1$, the set

$$\mathcal{B}_n = \{1, \sin(x), \ldots, \sin(nx), \cos(x), \ldots, \cos(nx)\} \tag{15}$$

consists of mutually orthogonal functions with respect to the standard L^2 inner product on $C[-\pi, \pi]$.

(b) Normalize the functions in $\mathcal{B}_n$ so that they have unit norm. Do the normalized functions form an orthonormal basis of $\operatorname{span} \mathcal{B}_n$?

(c) Find the orthogonal projection $T_n(x)$ of $f(x) = x$ onto $\operatorname{span} \mathcal{B}_n$. The orthogonal projection $T_n(x) = \operatorname{pr}_{\operatorname{span} \mathcal{B}_n} f(x)$ is called the *Fourier approximation* of $f(x)$. The series $\lim_{n\to\infty} T_n(x)$ is called the *Fourier series* of $f(x)$.

(d) Plot $f(x) = x$ together with its Fourier approximations $T_1(x), T_5(x)$, and $T_{25}(x)$ on the interval $[-\pi, \pi]$.

Solution:

[2] `https://youtu.be/HuHRkO9Y3Q4?si=4UHmqGdfh3c6dLA4`

(a) Let's compute all inner products between the functions in $\mathcal{B}_n$:

$$\begin{aligned}
\langle 1, \sin(kx)\rangle &= \int_{-\pi}^{\pi} \sin(kx)dx = -\frac{\cos(kx)}{k}\bigg|_{-\pi}^{\pi} = 0, \quad \text{for } k = 1, \dots, n, \\
\langle 1, \cos(kx)\rangle &= \int_{-\pi}^{\pi} \cos(kx)dx = \frac{\sin(kx)}{k}\bigg|_{-\pi}^{\pi} = 0, \quad \text{for } k = 1, \dots, n, \\
\langle \sin(kx), \cos(mx)\rangle &= \int_{-\pi}^{\pi} \sin(kx)\cos(mx)dx = \frac{1}{2}\int_{-\pi}^{\pi} \left(\sin((k+m)x) + \sin((k-m)x)\right) dx = 0, \quad \text{for } k \neq m, \\
\langle \sin(kx), \sin(mx)\rangle &= \int_{-\pi}^{\pi} \sin(kx)\sin(mx)dx = \frac{1}{2}\int_{-\pi}^{\pi} \left(\cos((k-m)x) - \cos((k+m)x)\right) dx = 0, \quad \text{for } k \neq m, \\
\langle \cos(kx), \cos(mx)\rangle &= \int_{-\pi}^{\pi} \cos(kx)\cos(mx)dx = \frac{1}{2}\int_{-\pi}^{\pi} \left(\cos((k-m)x) + \cos((k+m)x)\right) dx = 0, \quad \text{for } k \neq m.
\end{aligned} \tag{16}$$

(b) The L^2-norms of the functions in the set $\mathcal{B}_n$ are:

$$\begin{aligned}
\|1\| &= \sqrt{\int_{-\pi}^{\pi} 1dx} = \sqrt{2\pi}, \\
\|\sin(kx)\| &= \sqrt{\int_{-\pi}^{\pi} \sin(kx)^2 dx} = \sqrt{\pi}, \\
\|\cos(kx)\| &= \sqrt{\int_{-\pi}^{\pi} \cos(kx)^2 dx} = \sqrt{\pi}.
\end{aligned} \tag{17}$$

The normalized functions

$$\xi_0(x) = \frac{1}{\sqrt{2\pi}},\ \phi_1(x) = \frac{\sin(x)}{\sqrt{\pi}},\ \dots,\ \phi_n(x) = \frac{\sin(nx)}{\sqrt{\pi}},\ \psi_1(x) = \frac{\cos(x)}{\sqrt{\pi}},\ \dots,\ \psi_n(x) = \frac{\cos(nx)}{\sqrt{\pi}} \tag{18}$$

form an orthonormal basis of $\operatorname{span}\mathcal{B}_n$, since they span $\operatorname{span}\mathcal{B}_n$ and are linearly independent because of mutual orthogonality.

(c) The Fourier approximation of $f(x) = x$ is given by

$$T_n(x) = \operatorname{pr}_{\operatorname{span}\mathcal{B}_n} f(x) = \alpha_0\xi_0(x) + \sum_{k=1}^{n} \left(\beta_k\phi_k(x) + \gamma_k\psi_k(x)\right), \tag{19}$$

where (using integration by parts in the second line and symmetry in the first and third)

$$\begin{aligned}
\alpha_0 &= \langle f, \xi_0\rangle = \int_{-\pi}^{\pi} \frac{x}{\sqrt{2\pi}}dx = 0, \\
\beta_k &= \langle f, \phi_k\rangle = \int_{-\pi}^{\pi} \frac{x\sin(kx)}{\sqrt{\pi}}dx = -\frac{2\sqrt{\pi}(-1)^k}{k}, \\
\gamma_k &= \langle f, \psi_k\rangle = \int_{-\pi}^{\pi} \frac{x\cos(kx)}{\sqrt{\pi}}dx = 0
\end{aligned} \tag{20}$$

So, the Fourier approximation of $f(x) = x$ is

$$T_n(x) = 2\sum_{k=1}^{n} \frac{(-1)^{k+1}\sin(kx)}{k}. \tag{21}$$

(d) Figure 1 shows the function $f(x) = x$ along with its Fourier approximations $T_n(x)$ for $n = 1, 5$ and 25. Notice the persistent overshoots and oscillations that occur near $-\pi$ and π. This is often referred to as the *Gibbs phenomenon*. This happens because the Fourier series converges to $f(x)$ pointwise almost everywhere, but not uniformly near discontinuities. In this example, $f(x)$ is continuous on $(-\pi, \pi)$, but has a jump discontinuity at $-\pi$ and π when $f(x)$ is extended to $\mathbb{R}$ periodically.

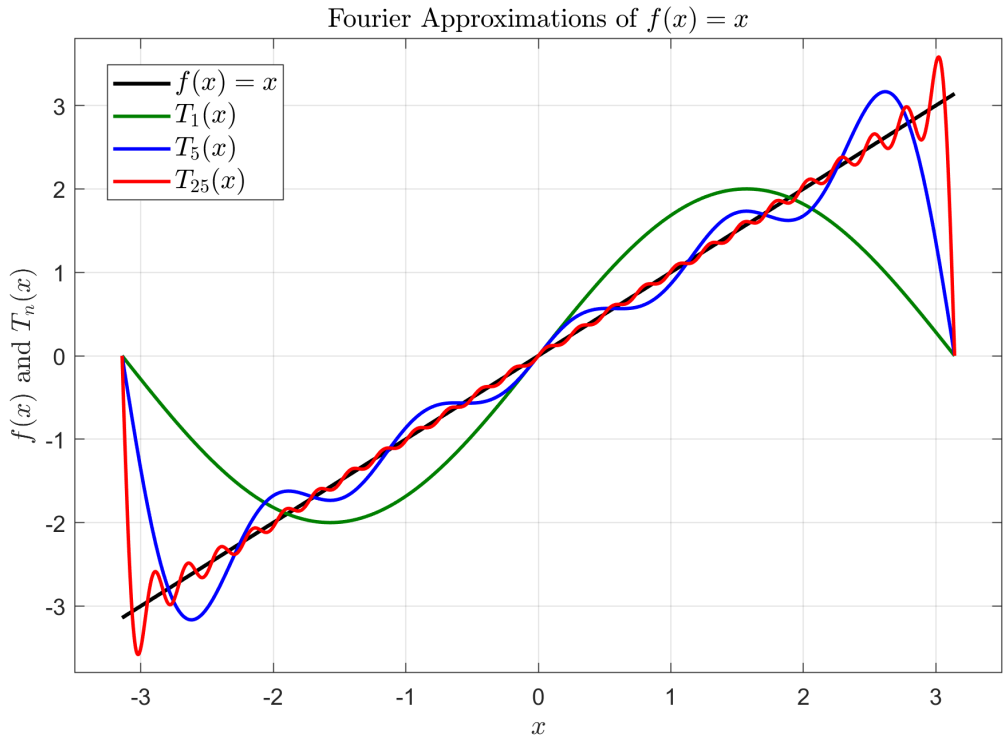

FIGURE 1. The function $f(x) = x$ and its Fourier approximations.